

VD Detector Protection System

Status and work towards a DPS conceptual design

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I&I meeting — November 2026

The hardware connection list is still growing

Known starting points

Rack power distribution units (PDUs).

The TPC electronics / Bottom Drift Electronics (BDE) protection design and its test work.

Work still needed

A complete list of hardware inputs, outputs and feedback between the equipment and the DPS Programable Logic Controller.

A conceptual design covering all VD DPS connections.

Approach: extend the existing TPC/BDE protection schema based on a central PLC.

Target: gather the consortia and LBNF connection information by late January 2027.

DPS protects equipment; SC operates and records

DPS	Detector Protection System: arbitrates protection, applies permits and inhibits, and checks the response.
DSCS	DUNE Detector Slow Controls System (SC): operates and monitors equipment, coordinates requests and keeps the logs.
DAQ	Data acquisition: configures readout and records detector data.
BMS / Cryo / SCADA	Building management, SCADA Controls and Cryogenics: operate their systems and report their state to SC.
Consortia	Define operating limits, protection actions and allowed response times.
Local interlocks	Can trip equipment without SC or the central DPS.

Protection trips take priority over operating requests.

SDSTA issues personnel-safety alarms.

Facility interfaces may connect to SC or the DPS PLC

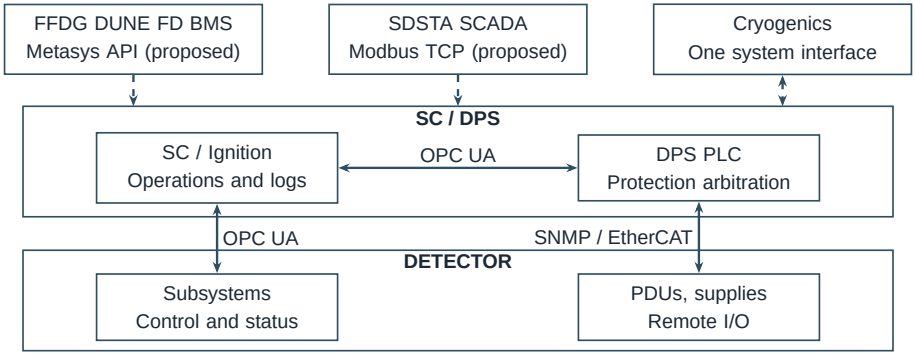


Diagram. Dashed: facility links under study. Cryogenics exchanges information in both directions.

The receiving endpoint may be SC/Ignition, the DPS PLC, or both; it is still to be agreed.

SC ↔ DPS: OPC UA. DPS ↔ equipment: SNMP / EtherCAT.

Build the DPS brain on the BDE PLC architecture

- Industrial off-the-shelf hardware; modular I/O supports expansion and scaling.
- Vendor support for maintenance and upgrades.
- Reuse the BDE panels across the ∼70 m mezzanine; extend to other systems.

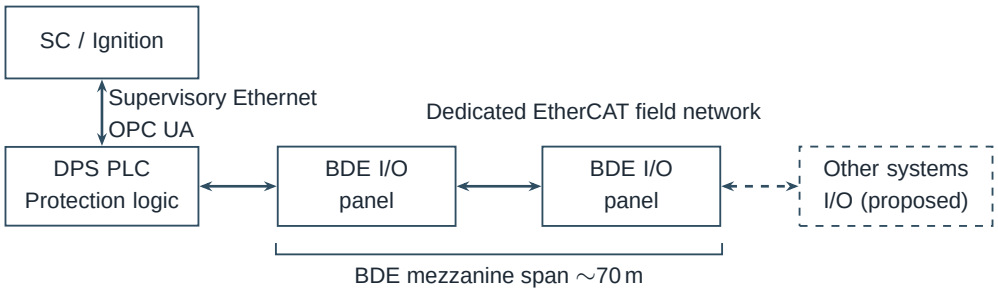
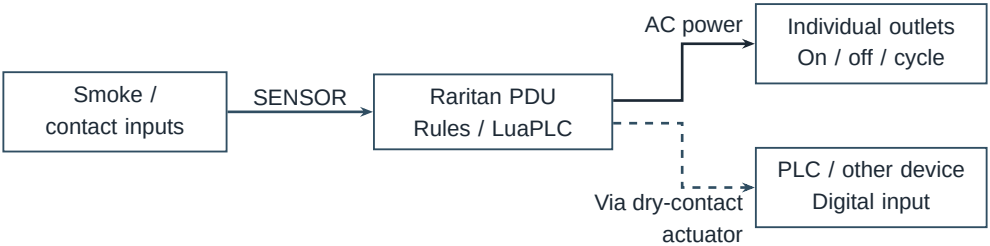


Diagram. Logical sketch. EtherCAT field network is separate from supervisory Ethernet. Dashed: proposed extension.

Raritan PDU controls on dune-fd-test01



Capability	Demonstration
Single outlets	On / off / cycle a test load; check outlet state and power.
Contact output	Send a fault/permit bit to a PLC or other device using an actuator.
Local logic	Rules: contact/threshold → output. LuaPLC: delays, sequences, reset.

Cryogenics presents one system to DUNE

The cryogenic controls are in the Central Utility Cavern (CUC). Their internal divisions appear as one system to DUNE FD DSCS.

Function	Detector interface
Process control	Cryogenics operates and protects its own process.
Protection inputs	Liquid-argon level, relief-valve state and boiling-risk conditions can affect HV and bias permits.
Information path	Cryogenics ↔ SC and/or DPS; endpoint to agree. Trace each measurement to its actual sensor.
Before operation	Agree readiness states, limits and the response to missing or invalid data.

The detector running conditions are still being agreed.

Trace each interlock from its source to the response

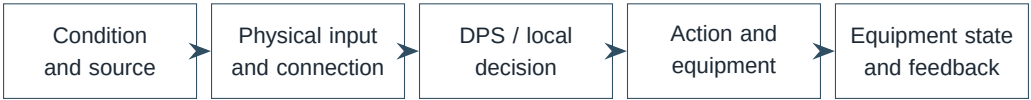


Diagram. Review each interlock from its source and physical input through the decision, action and equipment response.

Start with the interlocks proposed by the consortia in their interface documents (ICDs).

For each one, identify the existing hardware, missing connections and required additions. Check the time from detection to completed shutdown.

Use the rack lists, drawings, parts lists and power calculations.

Each group must identify its hardware connections

For every connection	Record
Source and destination	Equipment, rack / panel, connector, channel and drawing revision
Signal and interface	Measurement or state; units, normal / fault values; wired I/O or network protocol
Protection action	Condition, affected load, permit / inhibit and allowed response time
Feedback and recovery	Equipment response; lost-signal behaviour; reset and re-enable
Responsibility and gaps	Consortium or LBNF contact; evidence, missing items and next action

Use one common register for the consortia and the LBNF facility. Each group confirms that its list covers its hardware.

A trip removes the permit; recovery needs a new request



Allow: the request is valid, required conditions are met and no trip is active.

Trip: remove the permit, retain the cause and check the equipment response. SC logs the event.

Recover: reset the trip, then request a separate re-enable.

Diagram. Proposed controller: a commercial programmable logic controller (PLC) based on BDE, located in the CUC. Loss-of-signal behaviour is set for each protected load.

BDE is nearly complete; PDS has two decisions

Critical interlock	Status and group action
BDE cooling Fan fault → affected LV off.	**Nearly complete.** Close the remaining checks, including whether associated bias must also shut down, and in what order.
PDS / purity monitors Flash-lamp activity → bias inhibit or data tagging?	**Action — PDS + CalCI:** assess whether SiPM protection requires inhibition, or whether tagging the affected data is sufficient.
PDS power over fibre (PoF) Fault → inhibit affected PoF power.	**Action — PDS:** define the inhibition path, affected boxes, response time and feedback. PDU switching and box-enable are candidates.

Red marks the decisions needing group input. PoF inhibition and the purity-monitor rule remain open.

Agree the CalCI, HV and TDE protection interfaces

Critical interlock	Group action
CalCI laser / PDS Conflicting requests → block request.	**CalCI + PDS:** agree priority and emission-off or shutter-closed confirmation before PDS returns.
HVS Low LAr → affected HV/bias off. Drift HV off → associated field-cage termination off.	**HVS + Cryogenics / CalCI:** map level, relief-valve and boiling-risk signals. Agree supply groups, limits and shutdown order.
TDE / rack protection Rack fault → affected PDU outlets off.	**TDE + DPS:** define trip conditions, outlets, timing and feedback. Keep local crate protection independent.

CalCI: Calibration and Cryogenic Instrumentation. HVS: High Voltage System. Red marks group actions.

Use Batavia and SURF benches to test the protection paths

Batavia — HEERC G271

Reported progress

Initial BDE demonstrator design completed and reviewed with the WIEC team at BNL.

To do

Apply BNL feedback, test fan tachometer readout, then assemble and install the demonstrator.

SURF — Nevada room

`dune-fd-test01`

Available

PDU, smoke sensors, relay and GIZMo; Ignition, OPC UA bridges and emulators.

To do

Expand the setup along the Batavia model, adding the PLC/I/O and subsystem interfaces.

Test on both benches: faults, inhibits, equipment feedback and response time; lost communications, reset and re-enable.

Gather inputs by January; complete the design by April

Late January 2027

Gather the information

Consortia and LBNF return their connections, limits and required actions.

Include feedback, response times and lost-signal behaviour.

Confirm coverage; name contacts and remaining gaps.

Mid-April 2027

Complete a comprehensive DPS design

Architecture, controller and I/O allocation; interfaces and protection rules.

Fault response, shutdown order and recovery.

Drawings, parts, power needs and test plan; record open choices for review.

Work in parallel: use the Batavia and SURF benches to check representative protection paths as the design develops.

Planning targets. Hardware qualification and installation are separate steps.



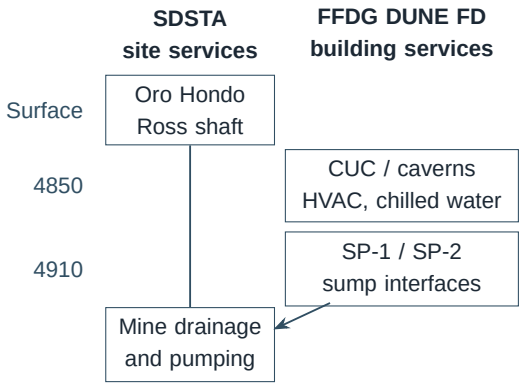
Backups

Agree responsibility and control-engineering support

Proposed responsibility	Work
Manuel + control engineer	VD hardware design, SURF interfaces and procurement
Terri and Trevor	BDE panels, PLC and I/O implementation, bench tests
Geoff + Manuel	SC commands, DPS arbitration and recovery
Vivian + consortia / DPS	Equipment requirements and interface-document updates

Support needed: control-engineering time and access to the existing test stands.

The FFDG BMS connection needs an API licence



Proposed connection

ISD proposes a Metasys API in Batavia to connect the FFDG DUNE FD BMS to SC.

Tested with SBN building data

The API and OPC UA publication have been exercised.

Needed for live data

The Metasys API licence.

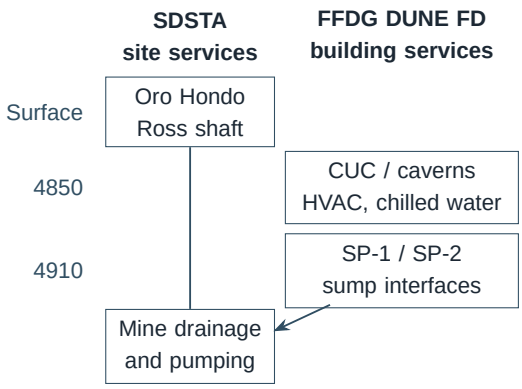
Diagram. Service boundaries at SURF. The map shows the scope of the systems, not individual controllers.

How FFDG BSI status can support detector protection

Reported in discussion	What we need to resolve
No 24-hour support; issues attended next business day	Decide a schema of human informed decisions using this and what happens when it is unavailable.
Dedicated sensors recommended for critical equipment protection	Identify where detector-owned sensors are needed.

BMS access alone does not establish a protection interface.

The SDSTA SCADA interface needs a hardware survey



Hardware survey underway

Controller, connection and network ports at the Ross Dry.

Protocols under investigation

Modbus TCP or OPC-UA could be options.

Agreement under discussion

Set of signals to use, faults and maintenance.

Diagram. SDSTA services extend beyond the detector caverns. The survey will establish the available points.

SDSTA proposes a restricted SCADA status interface

Topic	Reported proposal / next input
Interface	Modbus TCP, exposing a set of variables on selected ports. SDSTA retains control of its systems.
Write access	Status-only intent. Agree and check that the interface rejects writes and commands.
Variables	A list is pending, including Oro Hondo status and driver variables + HVAC.

Still to agree: variable meaning, data quality, loss-of-data behaviour and network access.



Detector

Ten BDE groups use the existing bottom-drift design

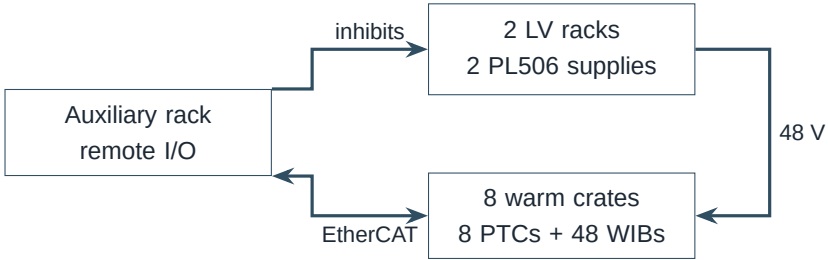


Diagram. One documented BDE group. The supply inhibit and 48 V path are separate from the two-way EtherCAT I/O link.

Ten groups: 10 auxiliary racks, 20 low-voltage racks and 20 PL506 supplies; 80 warm interface electronics crates (WIECs), containing 480 interface boards (WIBs).

Each crate has a power and timing card (PTC). Ten bias racks provide the separate charge-readout-plane and field-cage interfaces.

Available: panel drawings, parts lists, power calculations and Ignition screens.

Extend the BDE approach into a full DPS concept

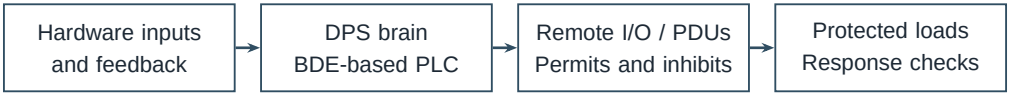


Diagram. Proposed extension of the existing TPC/BDE protection approach. Controller and I/O sizing follow the complete connection list.

Reuse the PLC, remote-I/O and protection-rule approach where it fits. Map the additional subsystem and facility interfaces.

SC carries operating requests and records equipment states and trips.

Test each added path from the fault input to confirmed equipment response.

A cooling fault removes power from the affected crate

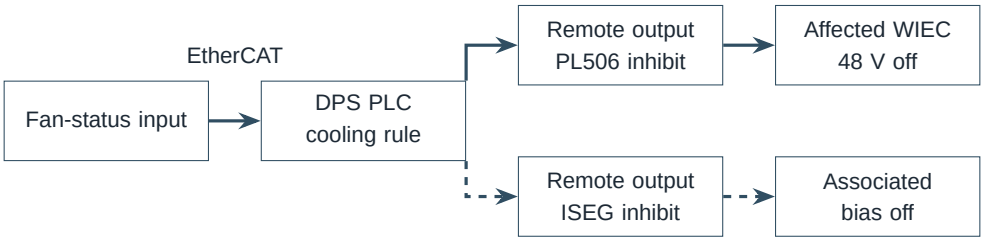


Diagram. Solid: documented fan-to-supply inhibit. Dashed: associated bias shutdown, if required by the VD rule.

A cooling fault removes power from the affected crate. DPS checks the supply state and voltage; SC records the trip.

The dependency and shutdown order for associated bias are still being agreed.

SC and DAQ use distinct DAPHNE service paths

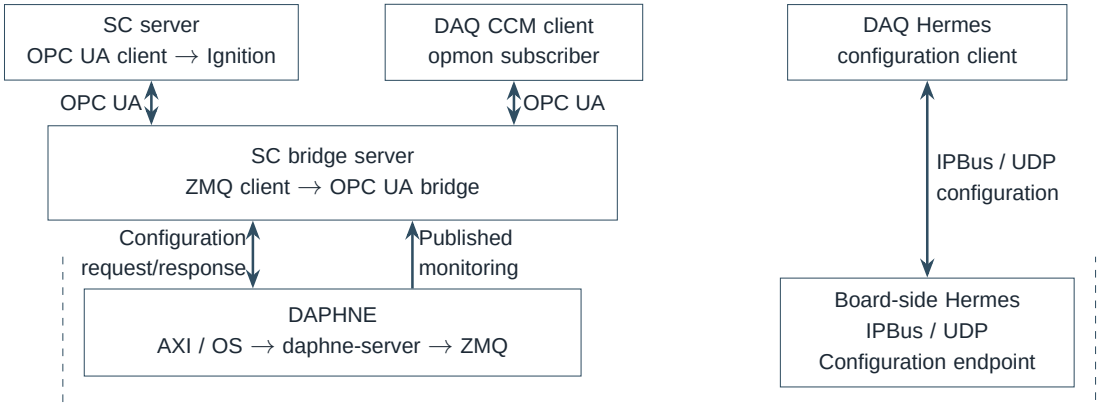


Diagram. Working SC–PDS–DPS ICD, Fig. 1. PDS: 34 LV feeds; cathode devices use PoF (next slide).

Proposed SiPM bias inhibit: DPS → SC → DAPHNE; SC returns bias status and informs DAQ.

PDS must define the PoF inhibition path

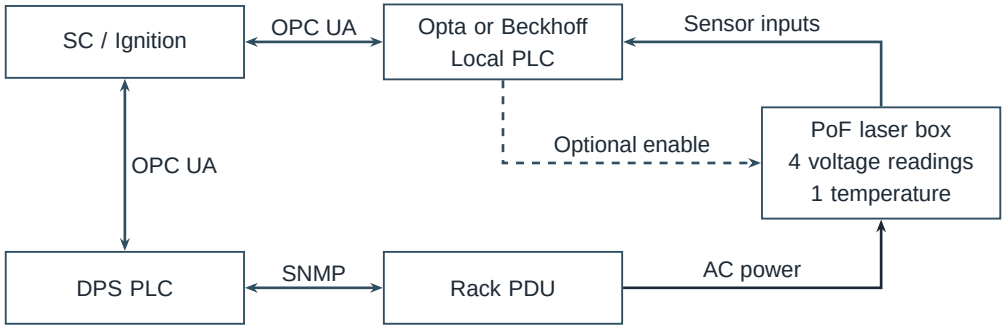


Diagram. Candidate paths: PDU power removal or a box-enable input, if available. Sensor inputs reach the local PLC.

80 boxes: 320 voltage + 80 temperature readings = 400 readings.

PDS must define how DPS inhibits power over fibre (PoF). Agree the actuator, affected boxes, response time, feedback and recovery.

LV supplies power DAPHNE and PDS calibration boxes

Working PDS LV scope: 34 feeds.

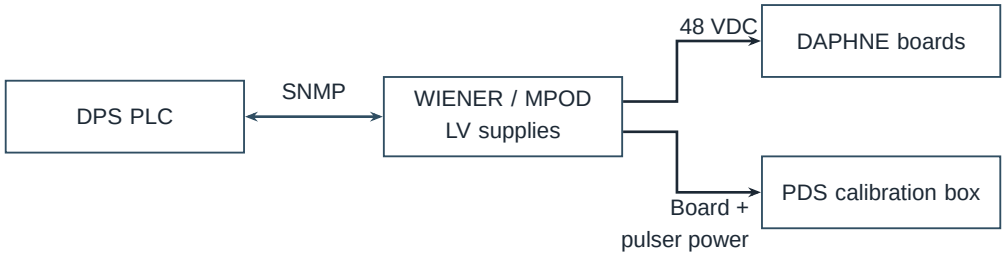


Diagram. Proposed SNMP control/status path. Power arrows group LV feeds; hardware-inhibit wiring remains open.

To complete: map loads to channels and inhibit groups; agree trip conditions and recovery.

Check supply state, voltage and current after each trip.

Drift HV and bias supplies have separate inhibit paths

One drift-HV supply; ten bias crates with 576 channels.

480 channels serve the charge readout planes (CRPs); 96 serve the field cage.

Equipment	Control and monitoring	Protection
Heinzinger drift-HV supply	SC / OPC UA through LabVIEW and cRIO; analog control and readback	DPS remote I/O operates the wired interlock
ISEG bias supplies	DPS uses SNMP; SC receives status and logs	DPS remote I/O operates wired channel inhibits

Example: low liquid-argon level removes the drift-HV permit. A drift-HV trip also shuts down the associated field-cage termination supplies.

Trace each protection input to a temperature sensor

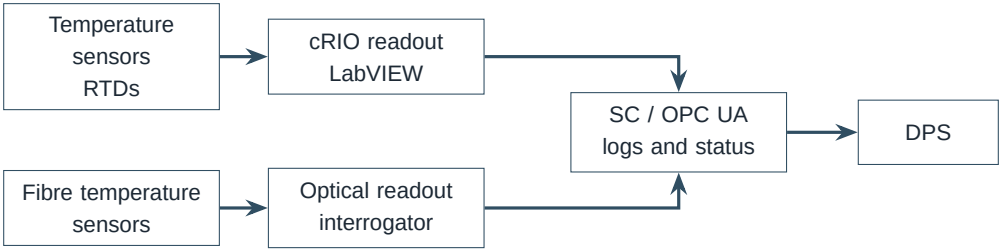


Diagram. RTD and fibre-sensor measurements reach SC through separate readout systems. Logical interfaces only.

Calibration and Cryogenic Instrumentation (CalCI) measures conditions inside the detector. Temperature profiles guide cooldown and operation.

For protection, each measurement needs a location, a limit, an allowed response time and a named piece of equipment to shut down.

Proposed laser rules block conflicting PDS requests

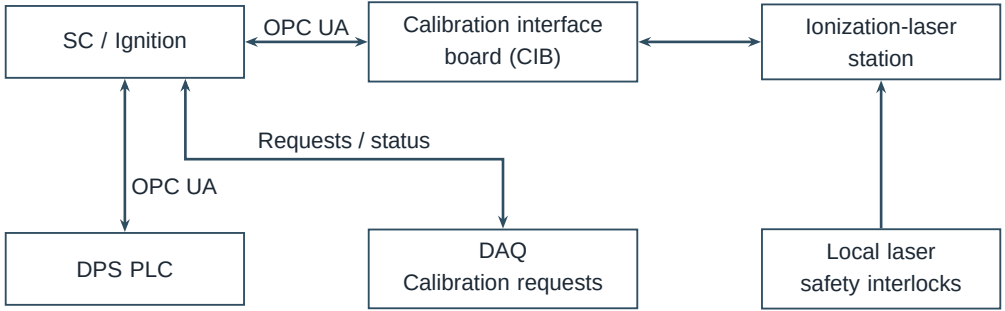


Diagram. SC exchanges requests and status with DPS, DAQ and the CIB. The local laser-safety path remains independent.

Four laser stations. Proposed rule: block conflicting laser and PDS requests. Restore PDS after confirmed emission-off or shutter closure.

Local laser-safety interlocks remain independent.

The purity-monitor inhibit rule still needs agreement



Diagram. Two-way links carry monitor control and status; SC sends activity tags to DAQ. The inhibit rule remains open.

Proposed approach: keep taking data and mark the intervals when the flash lamp is active.

Decision still needed: the older PDS interface document requires SiPM bias to be off during purity-monitor activity. Agree the rule before implementing the inhibit.

Measure the full path from fault to confirmed shutdown

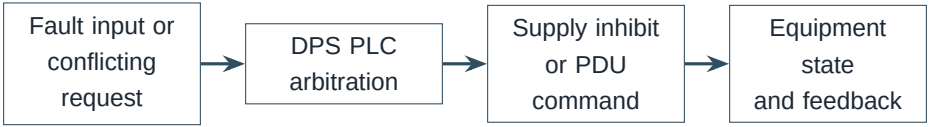


Diagram. Measure detection, arbitration, the output action and equipment feedback as one protection path.

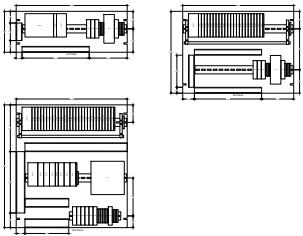
Use the existing BDE test setup to measure detection through completed shutdown.

Test local trips, lost communications, reset and re-enable. Check that SC logs the cause, decision and final equipment state.

No critical function requiring less than 1 ms has been identified so far. The consortium review and hardware tests will check this assumption.

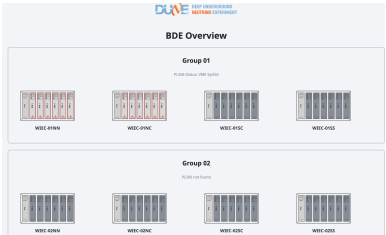
Existing panels and screens provide the BDE reference

TPC protection panel designs



Central PLC, bias I/O and LV/fan/heater reference panels.

BDE Ignition screens



Existing Ignition display of the BDE crate groups.

Parts list: CX5140 PLC, EK1100 couplers, EK1521 fibre links and EL2124 inhibit outputs.

Diagram. Reference panel design and existing operator screen. The VD group contains eight warm crates.

The fan trip removes the PL506 channel permit

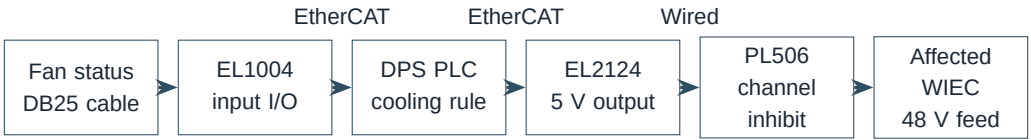


Diagram. Cooling fault → withdraw the permit for the affected PL506 channel.

Feedback: PL506 state and voltage/current → SNMP → DPS; OPC UA carries the result to SC for logging.

VD allocation check: six fan-status signals per WIEC means 48 per eight-WIEC group. Match these to the populated VD inputs and cables.

Six fans per crate is the design case used here. These are existing input requirements, not additional I/O.

DAQ CCM and Hermes use distinct DAPHNE services

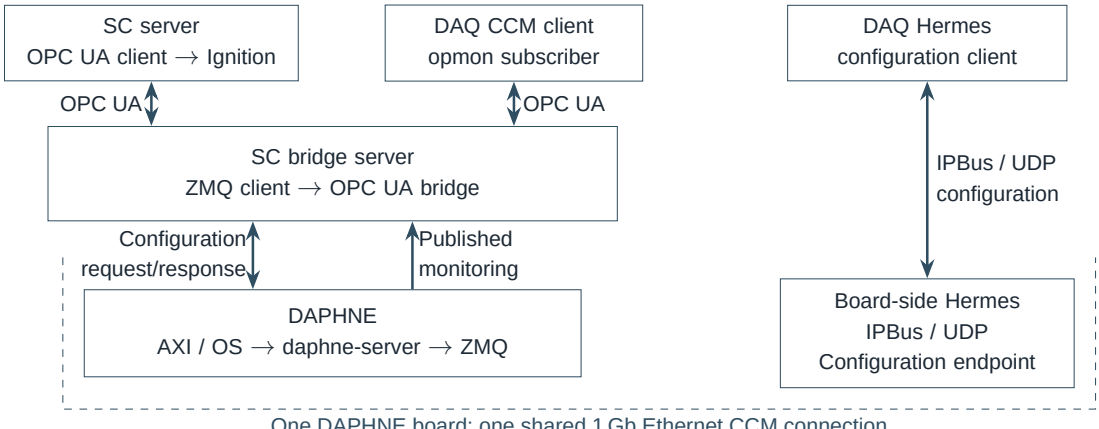


Diagram. Working SC–PDS–DPS ICD, Figure 1. DAQ CCM/opmon and Hermes remain distinct clients. The dashed bracket groups the board-side services.

Rack space is not a count of spare I/O terminals

Rack type	Count	DPS allocation and workbook space
BDE auxiliary	10	12U DPS; U3–U21 marked Space (19U); eight WIECs per rack
BDE low voltage	20	One PL506 per rack; four WIECs per supply
Bias high voltage	10	12U DPS; U3–U12 marked Space (10U)

Space marked in the rack workbook is not a new DPS reservation. Free DIN-rail length and I/O capacity depend on the populated VD panel.

U denotes rack height. The marked space is not an allocation of free PLC terminals.

The bias draft provides 1,440 outputs for review

Component	Per rail	10 racks
EK1100 EtherCAT coupler	1	20
EL9410 supply terminal	1	20
EL2124: four 5 V outputs	18	360
EL9011 end cap	1	20
24 V power supply	1	20

Each rail occupies 337 mm. The draft provides 1,440 output channels; the working VD inventory lists 576 bias channels.

Draft layout: the difference between 1,440 outputs and 576 bias channels is not verified spare capacity.

Supply capacity differs from expected power use

Equipment	Power / capacity	Basis and limit
PL506 input	2.25 kW each	20 supplies: about 45.1 kW expected AC input
WIEC fans only	380 W per group	$8 \times 6 \times 0.33\,\text{A} \times 24\,\text{V}$; excludes heaters and I/O
Bias DPS supply allowance	24 V, 40 A	960 W DC capacity per rack in the workbook

The inherited 831 W auxiliary figure needs a VD component budget covering fans, heaters and populated I/O.

Fan power assumes six fans per crate. Supply capacity is distinct from expected consumption.

Map each rack package against the existing allowance

Rack component	Reference hardware / allocation
PDU	Raritan PX3; inlet and outlet allocation for the connected loads
Smoke detector	4WTAR-B and EOLR-1 end-of-line relay
Contact interface	DX-PD2C5; 12 V supply to the smoke detector
Humidity	Record the probe and peripheral interface for each rack
Temperature / water leak	PDU peripheral sensors where fitted

The May workbook includes a 76-entry rack-protection/PDU allowance. Map the actual rack packages against this existing allowance.

Smoke/contact parts are retained as reference hardware; fire systems are outside DPS scope. Humidity-trip requirements remain open.

Local PDU trips need deliberate power restoration

PX3 setup guide	Local behaviour documented
Smoke, water, temperature	Local event rules and scripts can switch off the protected outlets
Network-switch outlet	Kept powered during the guide’s fallback shutdown
After a local trip	Power is restored deliberately; no automatic restoration

The VD control chain is Ignition → DPS PLC → PDU. Local trip status returns to the PLC and to SC for logging.

The reference prototype used direct Ignition control and included smoke contacts. Fire systems remain outside DPS scope; the VD PLC path needs its own communication-loss and recovery tests.

Keep subsystem quantities separate from crate totals

Subsystem	Working inventory and interfaces
PDS readout / calibration	34 LV feeds in the current presentation scope; supply status and control interfaces
Power over fibre (PoF)	80 boxes; 320 voltage and 80 temperature readings; Opta PLC or Beckhoff PLC alternative
TDE	20 mezzanine racks; DPS controls PDUs; crate-status and external-inhibit interfaces remain options
Calibration / cryogenic instrumentation	Level and temperature readout, purity monitors, ionization laser and neutron source

TDE and DAQ/timing contain 336 microTCA crates in total. Crate totals do not establish the number of PDUs or network switches.

Survey the 20 panel references to find installed hardware

Drawing area	Interfaces on the inspection maps
CUC, level 4850	Eight panel references: air handling, chilled water, power and associated equipment
Level 4910	North and south sump interfaces, SP-1 and SP-2
4850-08-ES group	Spare panel for future equipment; mapped to the access-drift inspection area
Other 4850 areas	Chillers, pumps, ventilation and generator interfaces

Marked inspection maps cover the 4850 and 4910 levels.

These are drawing references. The underground survey will establish what is installed.

The workbook allocates 76 mezzanine rack positions

Rack type	Count	Rack type	Count
BDE low voltage	20	PDS low voltage	4
BDE auxiliary	10	Drift HV	2
TDE	20	Reserved drift HV	2
Bias HV	10	Spare / CalCI	8

76 numbered mezzanine positions: SEDM-01 to SEDM-76.

The worksheet also has a separate concept for 40 top-plate mini-racks.

May 2025 workbook. Rack positions include reserved and spare allocations; they are not counts of PLCs or I/O stations.

Count shared controls and rack devices once

SC and network
- Ignition and logging services
- Operator / engineering stations
- Switches, routers and firewalls

DPS
- Central PLC and 24 V supplies
- EtherCAT couplers / fibre links
- Digital / analog / temperature I/O
- Fan/heater power outputs
- Wired permit / inhibit outputs

Facility interfaces to SC

FFDG BMS
- Metasys / Batavia API
- Building controllers

SDSTA SCADA
- Controllers and I/O: survey

Cryogenics
- Process and safety PLCs
- Operator stations / instruments

Rack infrastructure
- Raritan PX3 PDUs
- Smoke detectors and relays
- Contact interfaces / supplies
- Humidity / temperature probes
- Water-leak sensors

PDU-connected equipment
- Subsystem supplies and crates
- PoF boxes
- Computers and other rack loads

Facility systems keep their own controls. Rack protection and shared I/O are counted once.

The 336-crate total spans TDE and DAQ/timing

336 microTCA crates in total across TDE and DAQ/timing.

Bottom Drift Electronics
- 10 groups; 20 PL506 supplies
- 80 warm crates
  - 80 power/timing cards; 480 WIBs
  - Fans, heaters and temperature inputs

High voltage
- 1 Heinzinger supply; cRIO / LabVIEW
- 10 ISEG bias crates
  - 576 channels: 480 CRP + 96 field cage
- 10 bias racks with remote I/O

microTCA crate equipment
- Carrier hubs (MCHs)
- Readout / timing modules
- Power units and cooling fans

Top Drift Electronics
- 20 mezzanine racks
- 20 LV supplies and distribution units
- Front-end boards / calibration units
- White Rabbit timing switches

DPS control of TDE is through the PDUs.

Keep PDS control and DAQ data links distinct

Photon Detection System

- 42 DAPHNE boards in the working inventory
- WIENER supplies: 34 LV feeds

Power over fibre

- 80 laser boxes
  - 320 voltage + 80 temperature readings
- Opta or Beckhoff monitoring PLCs
- PDU power; optional box-enable input

Light calibration and timing

- 8 light-calibration boxes
- Local optical timing fanouts

DAQ and timing

- DAQ computers and readout switches
- Timing / readout microTCA crates
  - Included in the 336-crate total
- GPS interface boards (GIBs)
- GPS-disciplined oscillator
- Timing masters and fanouts

DAQ data and timing use separate links. SC receives the agreed status and activity information.

Instrument counts and interfaces still need allocation

Temperature
- RTDs; readout crates / cards
- cRIO and LabVIEW stations
- Fibre sensors and interrogators

Liquid level
- CRP / field-cage sensors and readout boards
- Level transmitter and controller

Other instrument interfaces
- Cold cameras; GIZMO; impedance monitor
- Berghof / Siemens instrument controllers

Ionization lasers: 4 stations
- CIB and station controllers
- Power electronics and computers
- Shutters and local safety interlocks

Purity monitors
- HV/LV supplies and front-end electronics
- Flash lamps, lamp supplies and triggers
- Digitizers and LabVIEW computers

Pulsed neutron source
- Generator, controller and permit interface

The device list records the interfaces and quantities still awaiting allocation.

Use rack sensors and PDUs for shared protection

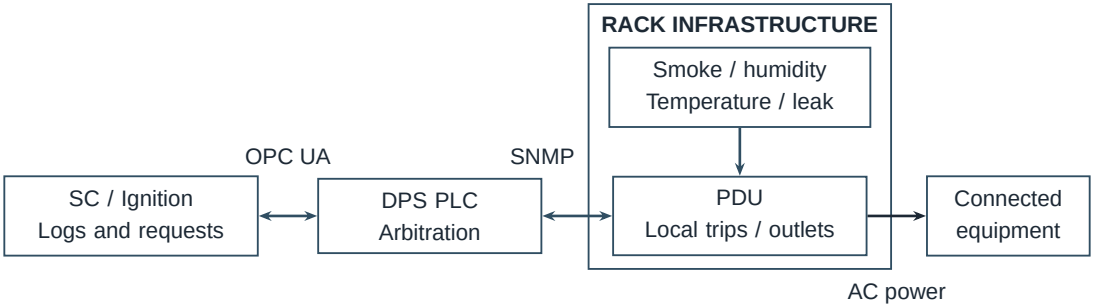
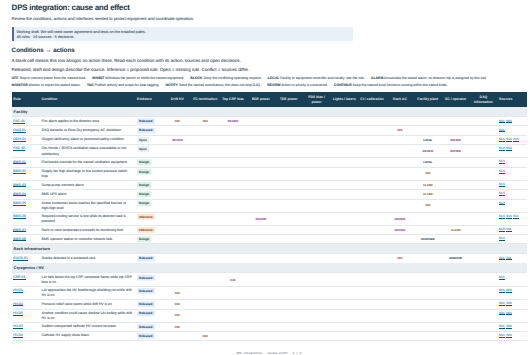


Diagram. Sensors feed the PDU; the outgoing line supplies AC power. Two-way links carry control and status.

Local PDU rules switch off affected outlets. Humidity is monitored; a humidity trip needs a separate requirement.

TDE + DAQ/timing: 336 microTCA crates. TDE uses 20 mezzanine racks. DPS controls their PDUs; crate-status and external-inhibit interfaces remain possible additions.

Full interlock matrix: conditions and actions



DUNE · WORKING MATRIX · 20.26-10-27 · V0.2

DPS integration: cause and effect

Review the conditions, actions and interfaces needed to protect equipment and coordinate operation.

Starting draft. We still need owner agreement and tests on the installed paths.
26 rules · 10 sources · 5 decisions

Conditions → actions

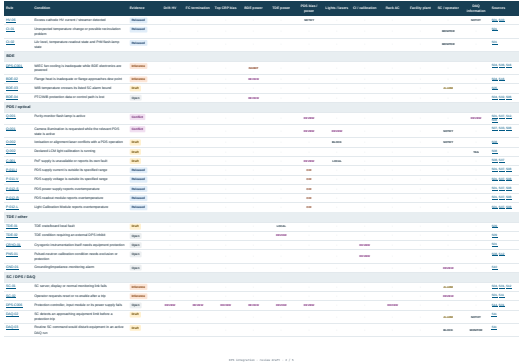
A blank cell means this row assigns no action there. Read each condition with its action, sources and open decisions.

Released, draft and design describe the source. Inference = proposed rule. Open = missing rule. Conflict = sources differ.

SET Beam or service power from the named load. **INHIBIT** Withdraws the permit or inhibit the named equipment. **BLOCK** Deny the conflicting operating request. **LOCAL** Facility or equipment-controller acts locally; see the rule. **ALARM** Annunciates the stated alarm; no distinct trip is assigned by this cell.
MONITOR Monitor or report the stated status. **TAG** Publish activity and scope for data tagging. **NOTIFY** Send the named event/status; this does not stop DAQ. **CONTINUE** Keep the named load functions running within the stated limits.

Rule	Condition	Evidence	Drift HV	FC termination	Top CRP bias	MDF power	TDF power	PHX blue+ power	Lights / lasers	CI / calibration	Rack AC	Facility plant	SC / operator	DAQ information	Sources
Facility															
FAC-01	Fire alarm applies to the detector area	Released	inhibit	inhibit	inhibit/alarm										DPS · FAC-01
FAC-02	DAQ transacts on Ross Dry emergency AC shutdown	Released									inhibit				DPS
FAC-03	Oxygen deficiency alarm or personnel-safety condition	Open	inhibit/alarm												DPS · FAC-03 · FAC-04
FAC-04	Gas Hookup / SCADA annunciation status unavailable or non-satisfactory	Open													DPS · FAC-04
MDF-01	Fire/smoke override for the named ventilation equipment	Design													MDF
MDF-02	Supply fan high discharge or low suction pressure switch trips	Design													MDF
MDF-03	Sump pump common alarm	Design													MDF
MDF-04	MMS UPS alarm	Design													MDF
MDF-05	Active condenser-basin reaches the specified low-low or high-high level	Design													MDF
MDF-06	Required cooling service is lost while its detector load is powered	Inference				inhibit/alarm		inhibit/alarm							DPS · MDF · FAC-01
MDF-07	Rack or room temperature exceeds its monitoring limit	Inference						inhibit/alarm							DPS · MDF
MDF-08	MMS operator station or controller network fails	Design													MDF
Rack infrastructure															
RACK-01	Smoke detected in a protected rack	Inference													DPS · RACK
Cryogenics / HV															
CRP-01	Lid fails below the top CRP composite frame while top CRP bias is on	Released													DPS
CRP-02	Lid approaches the HV feedthrough shielding/cone while top CRP HV is on	Released													DPS · CRP-02
CRP-03	Pressure relief valve opens while drift HV is on	Released													DPS · CRP-03
CRP-04	Another condition would cause shallow Lid-to-ceiling slide while HV is on	Released													DPS · CRP-04
CRP-05	Sudden unexpected cathode HV current increase	Released													DPS · CRP-05
CRP-06	Cathode HV supply shows down	Released													DPS · CRP-06

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Rule	Condition	Evidence	Drift HV	FC termination	Top CRP bias	MDF power	TDF power	PHX blue+ power	Lights / lasers	CI / calibration	Rack AC	Facility plant	SC / operator	DAQ information	Sources
PHX-01	Chiller cathode HV current / setpoint detected	Released						set/alarm						set/alarm	DPS · PHX-01
PHX-02	Unexplained temperature change or possible non-insulation pump failure	Released											acknowledge		DPS · PHX-02
PHX-03	Cold beam temperature-readout state and PHX flash-lamp state	Released											acknowledge		DPS · PHX-03
RIDE															
RIDE-01	MMS fan-cooling is inadequate while RIDE electronics are powered	Inference				inhibit									DPS · RIDE-01 · RIDE-02
RIDE-02	Flange heat is inadequate or flange approaches alert point	Inference				set/alarm									DPS · RIDE-02
RIDE-03	MMS temperature crosses its failed-SC alarm fanout	Draft											alarm		DPS · RIDE-03
RIDE-04	PTC/RIDE protection data-or-control path is lost	Open				set/alarm									DPS · RIDE-04 · RIDE-05
PDS / optical															
PDS-01	Pre-fly-monitor flash lamp is active	Inactive						inhibit/alarm						inhibit/alarm	DPS · PDS-01 · PDS-02 · PDS-03
PDS-02	Camera illumination is requested while the coherent PDS strain is active	Inactive						inhibit/alarm	inhibit/alarm				alarm/ack		DPS · PDS-02
PDS-03	Instrument or alignment laser monitors safe while a PDS operation	Draft							alarm/ack				alarm/ack		DPS · PDS-03
PDS-04	Declared LCM light-calibration is running	Draft												flag	DPS · PDS-04
PDS-05	PMT supply to unavailable or reports its own fault	Draft						set/alarm		inhibit/alarm					DPS · PDS-05
PDS-06	PDS supply current is outside its specified range	Released						inhibit							DPS · PDS-06
PDS-07	PDS supply voltage is outside its specified range	Released						inhibit							DPS · PDS-07
PDS-08	PDS power-supply reports overtemperature	Released						inhibit							DPS · PDS-08
PDS-09	PDS readout module reports overtemperature	Released						inhibit							DPS · PDS-09
PDS-10	Light Calibration Module reports overtemperature	Released						inhibit							DPS · PDS-10
TDE / other															
TDE-01	TDE overdrawn/load fault	Draft													DPS · TDE-01
TDE-02	TDE condition requiring an external CPS inhibit	Open						inhibit/alarm							DPS · TDE-02
ENVIRON	Cryogenic instrumentation fault needs equipment protection	Open													DPS · ENVIRON
PROT-01	Pulsed-resistor calibration condition needs exclusion or protection	Open													DPS · PROT-01
CAM-01	Ground/weld/spatter monitoring alarm	Open													DPS · CAM-01
SC / DPS / DAQ															
SC-01	SC server: display or thermal monitoring link fails	Inference													DPS · SC-01 · SC-02
SC-02	Operator requests reset or re-enable after a trip	Inference													DPS · SC-02
DAQ-01	Protection controller, input module or its power supply fails	Open													DPS · DAQ-01
DAQ-02	SC detects an approaching equipment-limit before a protection trip	Draft													DPS · DAQ-02
DAQ-03	Routine SC command would inhibit equipment in an active DAQ run	Draft													DPS · DAQ-03

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Diagram. Working matrix v0.2, 27 October 2026, conditions/action pages 1–2. The five-page companion also contains sources, decisions and readiness notes.

Use the interlock matrix to track each protection function

Working matrix v0.2: 46 rules. Owner agreement and tests remain open.

Condition	HV	BDE	PDS	Rack AC	DAQ	Evidence
Low LAr level	OFF	–	–	–	–	Released
BDE fan fault	–	INHIBIT	–	–	–	Inference
PDS overcurrent	–	–	OFF	–	–	Released
Rack smoke	–	–	–	OFF	–	Released
PrM lamp active	–	–	REVIEW	–	REVIEW	Conflict
LCM calibration	–	–	–	–	TAG	Draft

Excerpt: a dash assigns no action; REVIEW is unresolved; TAG records activity. Evidence describes the source, not approval or a passed test.

For each function: source, physical path, action, feedback, timing and reviewer.

Measure milestones with completed records and evidence

Proposed delivery checks

Delivery	Evidence to deliver	Progress measure
Connection collection	Confirmed inventory from each group: endpoints, signal, action, feedback, timing, contact and source.	Complete / required connection records, by consortium and LBNF.
Concept review	Functions mapped to hardware and logic; I/O gaps, failure/recovery behaviour, decisions and test plan reviewed.	Reviewed, mapped / in-scope functions; open decisions listed.

Hardware evidence is a separate gate: passed / planned tests; fault-to-final-state time against the agreed budget; loss, reset and recovery.

Agree scope and reviewers before counting. Targets: inputs by late January; design by mid-April 2027.

Prepare the hardware review

Work before review	Result
Trace each inhibit	Existing hardware, additions, cable routes and quantities
Test representative paths	Measured trip times, recovery and SC logging
Prepare purchases	Panel space, power, drawings, parts, licences, costs and spares

Order after the readiness review; install to match the site schedule.

Agree the facility-interface and full VD response-matrix review schedule with the consortia and LBNF.